Kunal Goenka

Data Scientist, Enterprise AI & Data at Boeing

k.goenka23@gmail.com linkedin.com/in/kunalgoenka23 github.com/itskunal23 Washington DC–Baltimore Area

AI engineer specializing in agentic systems, Model Context Protocol (MCP) integrations, and enterprise RAG across regulated environments like Azure Government Cloud. I take ambiguous operational bottlenecks from problem discovery to production software.

Experience

Data Scientist, Boeing

Jan 2026 to Present

Enterprise AI & Data (EAI&D), Arlington, Virginia

- Cut Nonconformance Report (NCR) engineering cycle time by 95%, from 80 to 4 hours per aircraft program each month, with an automated entity-resolution pipeline that combines heuristic rules with transformer-based semantic matching.
- Architected an enterprise agentic RAG platform on Model Context Protocol (MCP) tool servers, with hybrid retrieval (dense vectors and BM25), relational databases, multi-step query decomposition, schema-validated tool execution, and automated citation self-evaluation.
- Engineered an automated defense document ingestion pipeline on Amazon S3 that turns unstructured aerospace PDFs and legacy spreadsheets into structured, searchable vector metadata across 4,500+ knowledge objects, at sub-1.2-second P95 retrieval latency.
- Deployed citation-grounded semantic search across 3,000+ pages of airline safety manuals, with confidence-based routing that flags low-confidence predictions for human review and continuous improvement through active learning.
- Led the architecture and end-to-end development of an intern-recruiting platform on Azure Government Cloud for 630+ hiring managers, using Amazon Kiro's spec-driven agentic workflows to build PDF candidate parsing, hiring, conflict-of-interest compliance, interviews, rankings, offers, and Workday integration.
- Received the CTO's Performance & Innovation Award for the architecture and platform engineering behind the enterprise 2026 AI Summit, which reached 6.8K unique users and 24K page views.

Data Scientist, Boeing

Jul 2025 to Jan 2026

Employee Listening, Organizational Research & Assessments (ELORA), Charleston, South Carolina

- Engineered a multilingual NLP pipeline with spaCy and transformer-based models that processes 100K+ survey responses across 10+ languages, covering translation, named entity recognition, ethics classification, and thematic prediction, with F1 scores of 0.90–0.98 across the evaluated tasks.
- Architected a production semantic search engine on ChromaDB with custom embeddings, JWT authentication, and role-based access control, so executive leadership can query historical organizational data with citation-backed verification.
- Led a five-person engineering team that designed and shipped an AI Skill Gap Analysis MVP in under six weeks, combining unsupervised clustering with generative synthesis to show workforce capability gaps to HR and business leadership.
- Engineered the infrastructure for the 2025 Boeing AI Summit hub, which handled 10.9K+ page views across 3,400+ enterprise attendees.

Supplier Quality Intern, Boeing

Jun 2024 to Aug 2024

St. Louis, Missouri

- Built automated extraction and transformation pipelines across the Boeing Enterprise Data Warehouse and MySQL, querying 50K+ supplier performance records to replace manual spreadsheet validation.
- Standardized quality metrics and anomaly tracking across critical manufacturing suppliers and automated weekly executive reporting.

Technology Analyst, Virginia Tech Tech Fellows

Aug 2023 to Jan 2024

Blacksburg, Virginia

- Designed and deployed a full-stack restaurant recommendation web app (Flask, React) in 48 hours, owned its end-to-end architecture, and pitched the working prototype to Marriott corporate leadership.

Education

Georgia Institute of Technology

Aug 2026 to May 2028

Master of Science, Computer Science. Specializing in Artificial Intelligence (OMSCS). In progress.

Virginia Tech, Pamplin College of Business

Aug 2021 to May 2025

Bachelor of Science, Business Information Technology. Computer-Based Decision Support Systems. Graduated.

Skills

Agentic AI development, Retrieval-augmented generation (RAG), Python, Full-stack development, Model Context Protocol (MCP)

Agentic systems and retrieval

Model Context Protocol (MCP), LangGraph state machines, Hybrid retrieval (BM25 and dense vectors), Cross-encoder reranking, Multi-step query decomposition, Schema-validated tool execution, Citation self-evaluation, LLM-as-a-judge, Confidence-based human review

Language and documents

spaCy, Transformer models (RoBERTa), Named entity recognition, Text classification, Machine translation, Embeddings and ChromaDB, Entity resolution, Unsupervised clustering

Applications and services

Python, FastAPI, TypeScript, React, Java Spring Boot, Node.js, Flask, REST APIs

Cloud, data, and delivery

Azure Government Cloud, Azure SQL, Amazon S3, MySQL, JWT and role-based access control, Docker, CI/CD, Git, PyTest, OpenTelemetry, Tableau

Recognition

CTO's Performance & Innovation Award, Boeing

For the technical architecture and platform engineering behind the enterprise 2026 AI Summit: 6.8K unique users and 24K page views.

Activities

Virginia Tech Tech Fellows, Technology Analyst; Business Information Technology Club, Webmaster; Kaizen Technology Club, Board Member; Bollywood Club, Founder and President; Indian Students Association, Event Coordinator; Society of Indian Americans, Event Planner; Computational Modeling and Data Analytics Club, Member